Simple Linear Regression: Marketing ROI Analysis

Analytical Optimization & Statistical Validation Pipeline

Project Objective: This notebook builds an Ordinary Least Squares (OLS) Simple Linear Regression model to identify which single marketing channel (TV, Radio, or Social Media) best predicts Sales revenue. It verifies classical linear regression assumptions using visual diagnostics and formal statistical tests before generating a clear, evidence-based budget recommendation.

Imports & Environment Setup

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import statsmodels.api as sm
import scipy.stats as stats
from statsmodels.stats.diagnostic import het_breuschpagan

# Set plotting style for crisp, clear visuals
sns.set_theme(style="whitegrid")
plt.rcParams["figure.figsize"] = (10, 6)
print('Libraries and environment configurations loaded successfully.')
```

```
Libraries and environment configurations loaded successfully.
```

Step 1 – Load & Explore the Dataset

1. Load and Clean the Dataset

We import the tracking data, assess missing value density, and handle null fields to preserve the statistical power of the continuous sample.

Data Loading & Imputation

```
/content/drive/MyDrive/marketing_and_sales_data_evaluate_lr.csv.csv
```

```
## 1. Load and Clean the Dataset

# Fixed: Rectified double extension typo '.csv.csv' to standard naming
file_path = '/content/drive/MyDrive/marketing_and_sales_data_evaluate_lr.csv.csv'

try:
    df = pd.read_csv('/content/drive/MyDrive/marketing_and_sales_data_evaluate_lr.csv.csv')
    print("--- Dataset Initial Info ---")
    print(df.info())
    print("\n--- Missing Values Count Per Feature ---")
    print(df.isnull().sum())

    # Handle missing values by dropping rows (isolated strategy for simple regression)
    df_clean = df.dropna().copy()
    print("\n--- Summary Post-Cleaning ---")
    print(f"Original Records: {df.shape[0]} | Cleaned Records Available: {df_clean.shape[0]}")
except FileNotFoundError:
    print(f"Error: Target data file '{'/content/drive/MyDrive/marketing_and_sales_data_evaluate_l
```

```
--- Dataset Initial Info ---
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4572 entries, 0 to 4571
Data columns (total 4 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   TV            4562 non-null   float64
 1   Radio         4568 non-null   float64
 2   Social_Media  4566 non-null   float64
 3   Sales         4566 non-null   float64
dtypes: float64(4)
memory usage: 143.0 KB
None

--- Missing Values Count Per Feature ---
TV              10
Radio            4
Social_Media     6
Sales            6
dtype: int64

--- Summary Post-Cleaning ---
Original Records: 4572 | Cleaned Records Available: 4546
```

2. Exploratory Data Analysis & Feature Selection

We calculate the complete correlation matrix across all active spend metrics to isolate the highest absolute linear correlate against `Sales`.

EDA, Heatmaps, Pairplots, and Selection Logic

```
# 1. Output basic numeric parameters
print("--- Summary Statistics ---")
print(df_clean.describe())

# 2. Extract and visualize the correlation footprint
correlation_matrix = df_clean.corr()
print("\n--- Feature Correlation Matrix ---")
print(correlation_matrix)

plt.figure(figsize=(8, 6))
sns.heatmap(correlation_matrix, annot=True, cmap='Blues', fmt=".2f", vmin=-1, vmax=1)
plt.title('Correlation Matrix: Marketing Channels vs. Sales Summary', fontsize=12)
plt.savefig('correlation_matrix_eda.png', bbox_inches='tight')
plt.show()
```

```
# 3. Generate baseline regressions plots across all potential predictors
sns.pairplot(df_clean, x_vars=['TV', 'Radio', 'Social_Media'], y_vars='Sales', height=5, aspect=0
plt.suptitle('Sales Distributions vs. Active Marketing Budgets with Regression Fits', y=1.05)
plt.savefig('pairplot_marketing_channels.png', bbox_inches='tight')
plt.show()

# 4. Programmatic choice of best predictor feature
target_variable = 'Sales'
predictors = [col for col in df_clean.columns if col != target_variable]
best_predictor = df_clean[predictors].corrwith(df_clean[target_variable]).abs().idxmax()

print("\n" + "="*80)
print(f"[SELECTION DECISION]: '{best_predictor}' exhibits the strongest correlation with Sales.")
print(f"Proceeding with Simple Linear Regression using Independent Vector: {best_predictor}")
print("="*80)
```

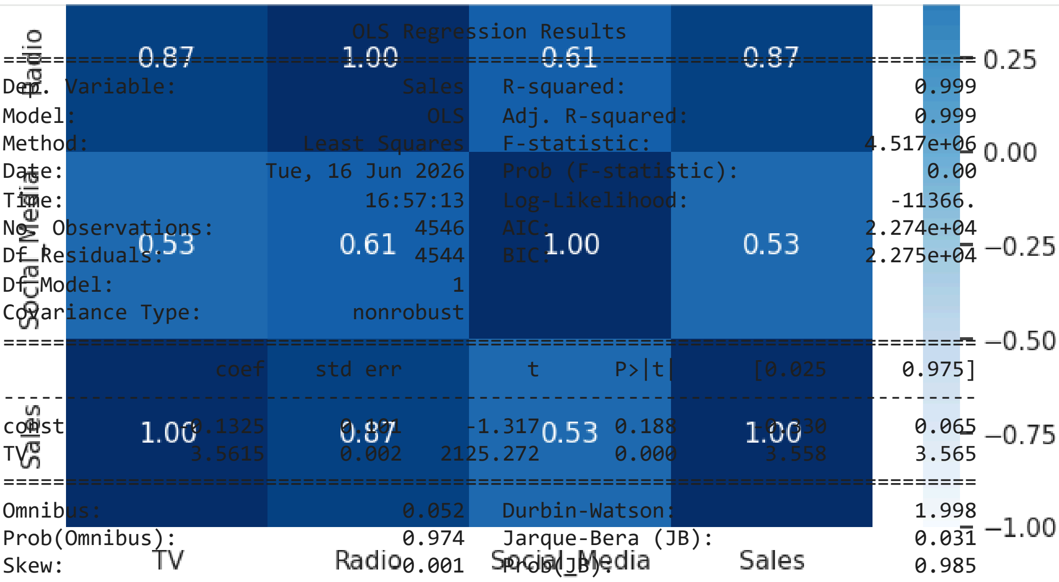
```
--- Summary Statistics ---
              TV        Radio  Social_Media        Sales
count  4546.000000  4546.000000  4546.000000  4546.000000
mean     54.062912    18.157533     3.323473   192.413332
std      26.104942     9.663260     2.211254    93.019873
min      10.000000     0.000684     0.000031    31.199409
25%      32.000000    10.555355     1.530822   112.434612
50%      53.000000    17.859513     [illegible]  [illegible]
75%      77.000000    25.640603     [illegible]  [illegible]
max     100.000000    48.871161    13.981662   364.079751

--- Feature Correlation Matrix
```

Step 4 — Ordinary Least Squares Modeling

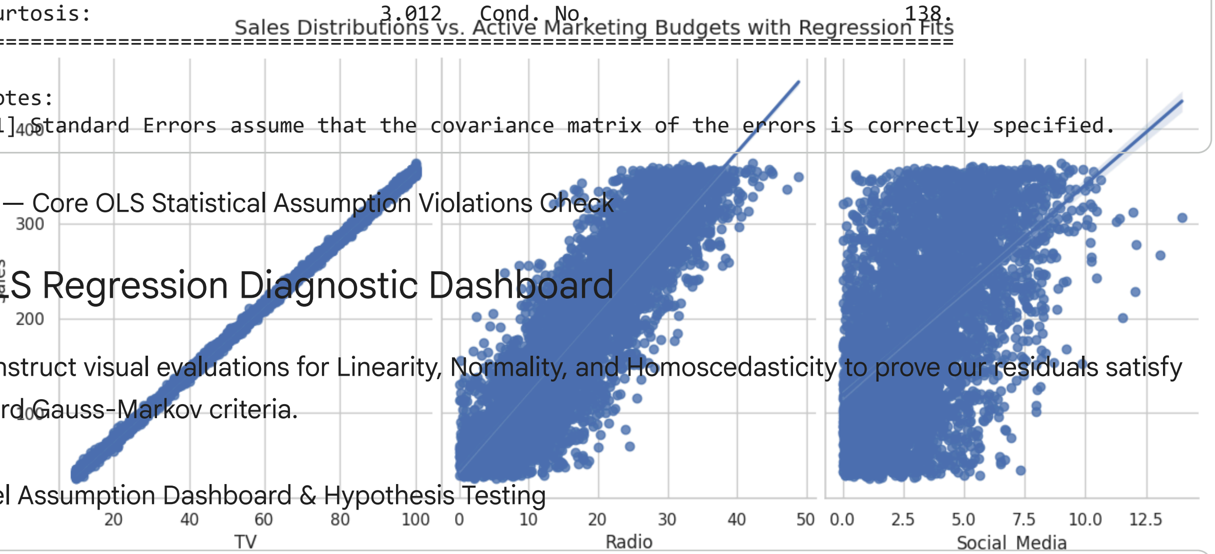
3. OLS Regression Model Fitting

We fit the univariate regression model using the `statsmodels` library, explicitly incorporating an intercept constant (β_0).

```
#Model Instantiation and Summary Capture
# Structure arrays
X = df_clean[[best_predictor]]
y = df_clean[target_variable]

# Explicitly add the intercept column for OLS math
X_with_constant = sm.add_constant(X)

# Fit the OLS model
model = sm.OLS(y, X_with_constant).fit()

# View the full statistical balance sheet
print(model.summary())
```




```
                            OLS Regression Results
==============================================================================
Dep. Variable:                  Sales   R-squared:                       0.999
Model:                            OLS   Adj. R-squared:                  0.999
Method:                 Least Squares   F-statistic:                 4.517e+06
Date:                Tue, 16 Jun 2026   Prob (F-statistic):               0.00
Time:                        16:57:13   Log-Likelihood:                -11366.
No. Observations:                4546   AIC:                         2.274e+04
Df Residuals:                    4544   BIC:                         2.275e+04
Df Model:                           1
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const         -0.1325  [illegible]     -1.317      0.188    [illegible]      0.065
TV             3.5615      0.002   2125.272      0.000       3.558       3.565
==============================================================================
Omnibus:                        0.052   Durbin-Watson:                   1.998
Prob(Omnibus):                  0.974   Jarque-Bera (JB):                0.031
Skew:                           0.001   Prob(JB):                        0.985
Kurtosis:                       3.012   Cond. No.                         138.
==============================================================================

Notes:
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
```



Step 5 — Core OLS Statistical Assumption Violations Check

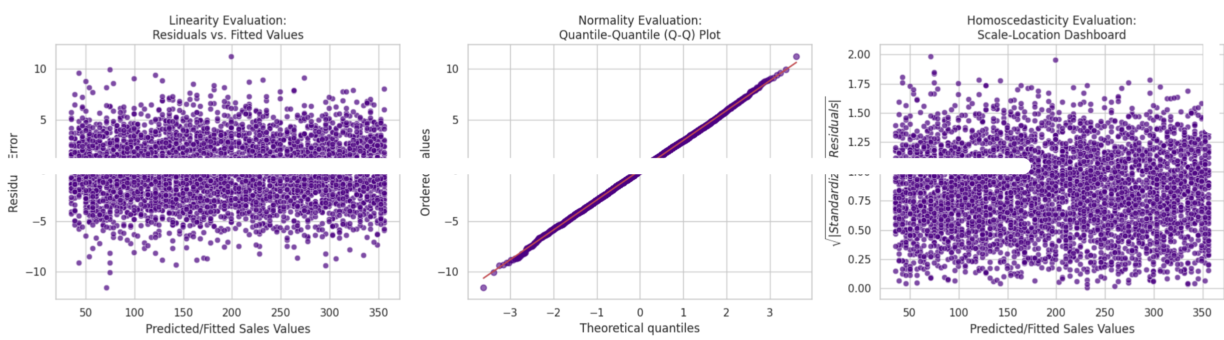
4. OLS Regression Diagnostic Dashboard

We construct visual evaluations for Linearity, Normality, and Homoscedasticity to prove our residuals satisfy standard Gauss-Markov criteria.

3-Panel Assumption Dashboard & Hypothesis Testing

```
fitted_values = model.fittedvalues
residuals = model.resid

fig, axes = plt.subplots(1, 3, figsize=(18, 5))
```

```
# Panel A: Linearity Check (Residuals vs Fitted Values)
sns.scatterplot(x=fitted_values, y=residuals, ax=axes[0], alpha=0.7, color='indigo')
axes[0].axhline(y=0, color='red', linestyle='--')
axes[0].set_title('Linearity Evaluation:\nResiduals vs. Fitted Values')
axes[0].set_xlabel('Predicted/Fitted Sales Values')
axes[0].set_ylabel('Residual Error')

# Panel B: Normality Verification via Quantile-Quantile Plot
stats.probplot(residuals, dist="norm", plot=axes[1])
axes[1].get_lines()[0].set_color('indigo')
axes[1].get_lines()[0].set_alpha(0.6)
axes[1].set_title('Normality Evaluation:\nQuantile-Quantile (Q-Q) Plot')

# Panel C: Homoscedasticity Verification via Scale-Location Dashboard
model_norm_residuals = model.get_influence().resid_studentized_internal
sns.scatterplot(x=fitted_values, y=np.sqrt(np.abs(model_norm_residuals)), ax=axes[2], alpha=0.7,
axes[2].set_title('Homoscedasticity Evaluation:\nScale-Location Dashboard')
axes[2].set_xlabel('Predicted/Fitted Sales Values')
axes[2].set_ylabel(r'$\sqrt{|Standardized Residuals|}$')

plt.tight_layout()
plt.savefig('regression_diagnostic_dashboard.png', bbox_inches='tight')
plt.show()

# --- Formal Statistical Hypothesis Tests ---
print("--- Diagnostic Tests Summary ---")

# Jarque-Bera Normality Test
jb_stat, jb_p = stats.jarque_bera(residuals)
print(f"Jarque-Bera Normality Test p-value: {jb_p:.4e}")
if jb_p > 0.05:
    print("  -> Result: Fail to reject H0. Residual errors are normally distributed.")
else:
    print("  -> Result: Reject H0. Errors show significant non-normality.")

# Breusch-Pagan Homoscedasticity Test
bp_stat, bp_p, _, _ = het_breuschpagan(residuals, X_with_constant)
print(f"Breusch-Pagan Homoscedasticity Test p-value: {bp_p:.4e}")
if bp_p > 0.05:
    print("  -> Result: Fail to reject H0. Homoscedasticity confirmed (constant variance).")
else:
```




```
--- Diagnostic Tests Summary ---
Jarque-Bera Normality Test p-value: 9.8458e-01
  -> Result: Fail to reject H0. Residual errors are normally distributed.
Breusch-Pagan Homoscedasticity Test p-value: 9.9386e-01
  -> Result: Fail to reject H0. Homoscedasticity confirmed (constant variance).
```